Overheated electrical joints: fire, danger to life and the cost of heat

Research report on the distribution-board joints photographed at the Chiang Rai site, with evidence, case studies, a before-and-after comparison and recommended practice

Site: Chiang Rai · **Photographs dated:** 1 September 2026 · **Monitoring data:** 28 September to 3 October 2026

Finding

More than one cable, including a twisted bundle, is clamped in one RCBO terminal. The joint has overheated: insulation is melted and copper is corroded. A second board shows the same habit.

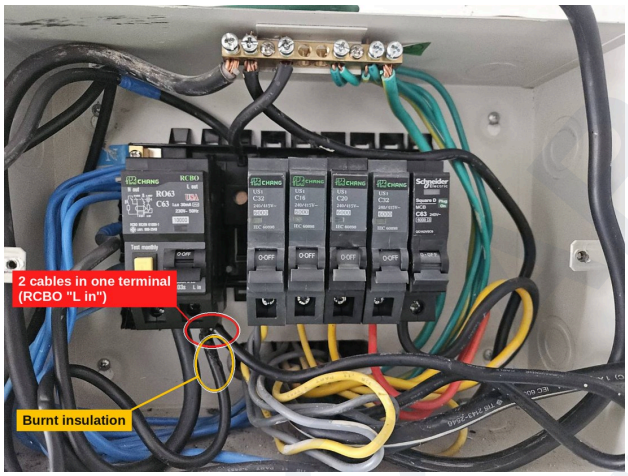
Danger

A hot joint is a source of ignition that no breaker detects. It also strips insulation from live conductors, so it threatens people by shock, smoke and fire.

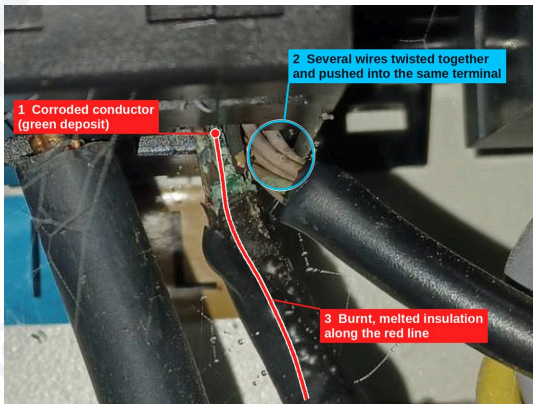
Cost and control

Heat at a joint is paid-for energy that does no work, and it is the first sign of larger losses and failures. Joints must be made correctly, tested and monitored under load.

1 What the photographs show



The board. The red circle marks the RCBO line-in terminal; the yellow outline marks scorched cable below it.

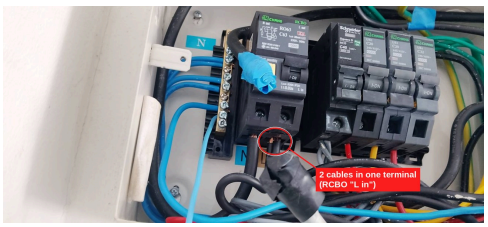


The terminal, enlarged. The numbers match the findings below.

1 Corroded conductor
Green deposit on the copper where it enters the terminal: a sign of heat, moisture and acid from overheated PVC.

2 Several wires in one terminal
Separate wires twisted into a bundle and forced in beside the main cable. The screw cannot grip them evenly.

3 Burnt, melted insulation
The damage runs down the cable from the terminal, which shows the heat started at the joint.

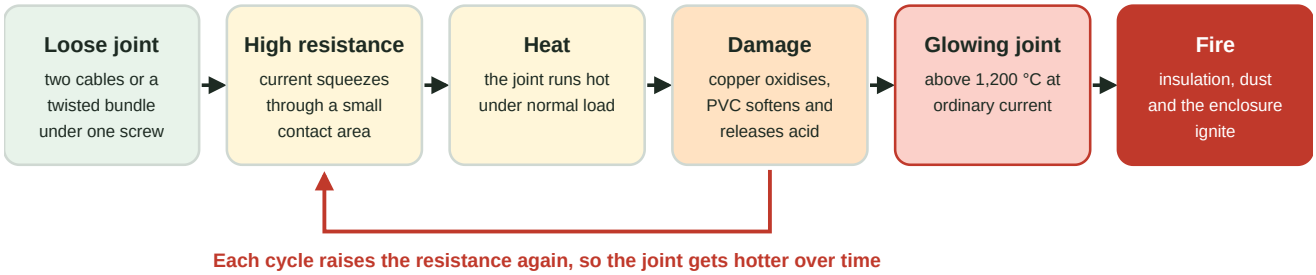


The same practice on a second board

Another board, judging by its different breakers, shows two conductors joined under tape and taken into one RCBO terminal. No burn is visible there yet. The fault looks like a working habit, not a one-off, so every board by the same installer should be opened and checked.

Also recorded at this site. The work was handed over with **no insulation resistance test**, **no thermal scan** and **wrong phasing**. These points are not visible in the photographs; Solar ESS recorded them on site.

2 The mechanism: how a loose terminal becomes a fire



Why several wires under one screw are loose

The screw bears on the outer strands only. Inner wires sit slack, and every heating and cooling cycle lets the bundle settle further. A twisted joint is also several circuits drawing current through one point.

Why the breaker does not trip

An MCB reacts to too much current and an RCBO to current leaking to earth. A hot joint causes neither: tests show it wastes only about 5 to 50 W, with normal load current flowing. The protection sees nothing wrong.

Why the copper went green

Overheated PVC releases hydrogen chloride. With moisture it forms acid that turns copper into green copper salts. The deposit conducts poorly, which raises the resistance again.

3 Evidence: how often this ends in fire

47%
of 1,745 fires in Bangkok in 2023–2025 were traced to electrical equipment (823 fires). Wall outlets and wiring alone accounted for 269.
Bangkok Metropolitan Administration, Dec 2025

69%
of US home fires involving electrical distribution and lighting equipment start in wiring and related equipment: about 31,960 fires and 400 deaths a year in total.
NFPA fact sheet, 2010–2014 data

5 times
more consumer-unit fires in London within five years. The main cause identified: high-resistance connections from incorrectly terminated cables.
London Fire Brigade, via Professional Electrician, 2016

1,200 °C
and above is reached by a glowing connection, at currents as low as 10 A. Many simply burn open; some ignite what is around them.
V. Babrauskas, EC&M, 2021

4 Case studies

Hospital switchboard fire, Shandong, China (2021)

The fire began at the bolted joint between a busbar and a circuit-breaker terminal. The bolts were 8 mm in 12 mm holes, so the contact carried about 40% of its rated capacity. The joint reached about 1,040 °C, the busbar sagged onto another phase, and the short circuit ignited the cabinet. It was detected early and no one was hurt.

Zhang et al., Fire, vol. 8 no. 11, 2025

Consumer-unit fires, London (2010s)

After the fire brigade traced a sharp rise in fires to badly terminated cables inside plastic consumer units, the UK wiring rules were changed: from 1 January 2016 consumer units in homes must have a non-combustible enclosure (BS 7671 regulation 421.1.201). The workmanship fault was common enough to force a change in the product.

Professional Electrician, 11 Feb 2016

Case studies, continued

Laboratory evidence on glowing connections

Tests on overheating connections measured only 5 to 50 W of heat and a 3 to 25 V drop at the joint, at ordinary currents such as 10 to 12.5 A, yet temperatures above 1,200 °C, hotter than the melting point of copper (1,083 °C). Many such joints simply burn open; some ignite what is around them.

V. Babrauskas, EC&M, 2021

Forensic study of failed connections (2022)

Microscopy of overheated connections found that a filament or pool of liquid copper oxide often carries the current at the hot spot. The paper reports a food-processing plant fire traced to such a joint, and an appliance joint that overheated without igniting its surroundings.

T. Korinek, Journal of the National Academy of Forensic Engineers, 39(1), 2022

This site is the same failure at an earlier stage

The joint has already overheated enough to melt insulation and corrode copper, but has not yet ignited. Fires like the two above develop through this stage.

5 Comparative study: danger of fire and danger to life

The same joint produces two different hazards. Fire damages property and stops production; the hazards to people act faster and need less energy.

	Danger of fire	Danger to human life
How it starts	The joint becomes a heat source inside the board, at normal load current.	Heat melts the insulation and leaves live copper exposed inside the board.
What is at risk	PVC insulation, dust, plastic enclosures, then cable routes and the building.	Anyone who opens the board or touches metalwork that has become live; everyone in the building if fire starts.
Why protection fails	An MCB needs overcurrent and an RCBO needs earth leakage. A hot joint gives neither.	The damage is on the line-in side of the RCBO, upstream of its sensing, so that RCBO cannot clear a fault there.
How little it takes	5 to 50 W at the joint is enough to exceed 1,200 °C.	RCDs for personal protection are rated 30 mA because larger currents through the body can stop the heart (IEC 60479).
Secondary effects	A sagging or carbonised conductor can short to another phase and ignite the cabinet (Shandong case).	Burning PVC gives dense smoke and hydrogen chloride gas. Smoke, not flame, is what usually kills in fires, often at night.
Made worse here by	The 57 kW machine: about 82 to 97 A per phase, so joint heat is far higher on machine days.	Wrong phasing: three-phase machines can run backwards; no insulation test, so hidden damage is unknown.
Warning before harm	Weeks to months of heating, visible to a thermal camera.	Often none: the first sign can be the shock itself.

Where this joint stands on the path to fire

Stage 1: Loose
No sign. Found only by a torque check.

Stage 2: Warm
Found only by thermal camera under load.

Stage 3: Discoloured
Melted insulation, green copper. **This site.**

Stage 4: Glowing
Above 1,200 °C. Smell, flicker, buzzing.

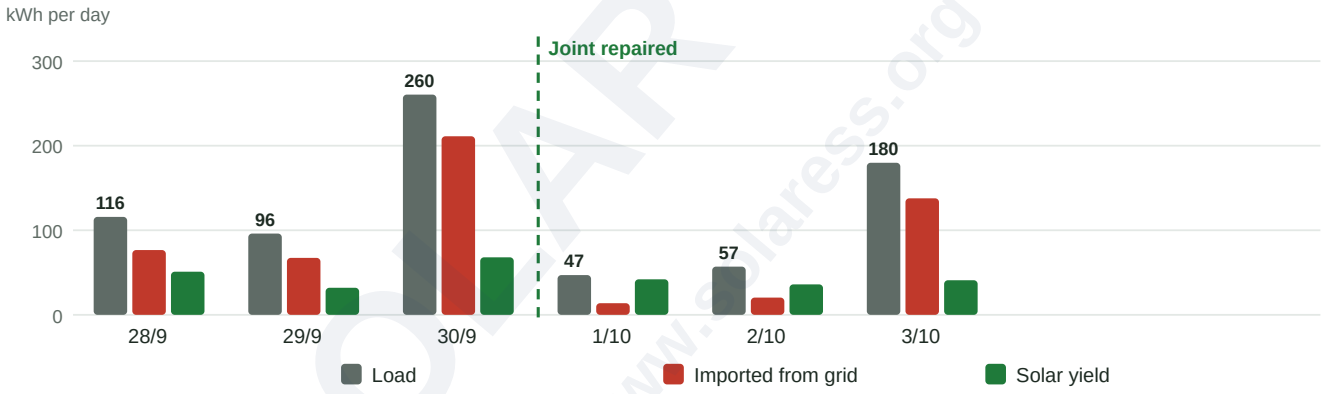
Stage 5: Fire
Insulation, dust and enclosure burn.

6 Comparison: before and after the joint was repaired

Condition	Before repair (as photographed)	After a correct repair	How to confirm
Contact	Several conductors under one screw; uneven grip	One conductor per terminal, ferrule, maker's torque	Torque check, pull test
Joint temperature	Hot enough to melt PVC and corrode copper	Within a few degrees of the neighbouring terminals	Thermal scan while the machine runs
Insulation	Melted and scorched along the cable	Damaged length cut out or cable replaced	Insulation resistance at 500 V DC, at least 1 MΩ
Shock risk	Bare live copper inside the board	No copper visible outside terminals	Visual inspection, cover fitted
Fire risk	Stage 3 of 5 and rising with every load cycle	Returned to that of a sound installation	Repeat scan after one month
Energy	Heat wasted at the joint; voltage drop to the load	Loss at the joint negligible	Voltage at board and at load, under load

The "after" column describes what a correct repair achieves. Solar ESS has no photographs or test records of the repaired joint in this report; the confirmations in the last column should be recorded on site.

7 Measured energy before and after the repair



Average per day

Per day	Before (28–30 Sep, 3 days)	After (1–3 Oct, 3 days)	Change
Load (consumption)	157.4 kWh	94.6 kWh	−40%
Imported from grid	118.4 kWh	57.3 kWh	−52%
Net import	103.8 kWh	52.0 kWh	−50%
Solar yield	50.3 kWh	39.6 kWh	−21%
Exported	14.6 kWh	5.3 kWh	−63%
Grid cost at 4.7 THB/kWh	556 THB	269 THB	−52%

The 57 kW machine and the high days

Day	Load	Extra over 50 kWh base	Equivalent hours at 57 kW
30/9	260.4 kWh	~210 kWh	~3.7 h
3/10	179.8 kWh	~130 kWh	~2.3 h
28/9	115.8 kWh	~66 kWh	~1.2 h
29/9	96.1 kWh	~46 kWh	~0.8 h

What the data supports

Comparing three days each side of the repair, average load fell from 157 to 95 kWh a day and grid import from 118 to 57 kWh a day: about 287 THB a day at 4.7 THB/kWh.

What it does not prove

Each period has one heavy machine day (260 kWh on 30 September, 180 kWh on 3 October) and two lighter days. Three days each side is a small sample, and the result follows how long the machine ran.

What would settle it

Compare days with the same machine hours before and after, using the hourly curve and a machine log, and record voltage at the load while it runs.

Hours assume full power; at part load the machine ran longer. The 50 kWh base is taken from the two quiet days (1 and 2 October). The repair date (between 30 September and 1 October) and the system size (about 20 kWp, no battery) are inferred from the data.

8 How heat raises the price of energy

Every hot joint is electricity bought and turned into heat

Heat at a joint = current² × joint resistance. The meter counts it exactly like useful load. Because the current is squared, the 57 kW machine multiplies the loss: nine times the current gives eighty-one times the heat.

Joint resistance	Heat at 10 A	Heat at 30 A	Heat at 90 A (machine running)
0.005 Ω (slightly loose)	0.5 W	4.5 W	~40 W
0.01 Ω	1 W	9 W	~80 W
0.05 Ω (badly corroded)	5 W	45 W	~400 W

1. Direct loss

A 400 W joint running 4 hours a day for 250 working days wastes about 400 kWh a year, about 1,900 THB, for one joint on one phase. A board has dozens of joints.

2. Heat feeds itself

Copper resistance rises about 0.4% per °C. A conductor running 50 °C hotter has about 20% more resistance, so it loses more and heats further.

3. Voltage drop

The volts lost at the joint are taken from the machine. Motors draw more current at low voltage and heaters run longer for the same work.

4. Peak demand

Where the tariff carries a demand charge per kW of monthly peak, losses and inrush from stalled restarts add to the peak that sets the charge.

5. Failure costs

A burnt breaker, cable or busbar means replacement parts, an emergency call-out and lost production while the machine stands idle.

In proportion

At this site the machine itself, not the joint, accounts for most of the bill (section 7). The joint adds a smaller, avoidable loss, and it is the one that can burn the building.

9 Practice: make connections carefully, then monitor them

Making the connection

- One conductor per terminal unless the maker identifies it for more (NEC 110.14(A))
- Ferrules on stranded conductors; no twisted or taped joints in the board
- Tighten to the maker's torque with a torque screwdriver, then pull-test
- Breaker and cable rated for the load: about 82 to 97 A per phase for the 57 kW machine
- Use a busbar or rated terminal block to split circuits
- Seal cable entries; fit covers

Testing before handover

- Insulation resistance at 500 V DC: at least 1 MΩ, and investigate any low value in a new installation
- Phase identification and phase sequence, with the board labelled
- RCD and RCBO trip test
- Thermal scan at working load, with images kept as the baseline
- Signed test record given to the owner

Monitoring in service

- Thermal scan at least once a year, and always while the machine runs
- Re-check torque one month after any new work
- Compare each joint with its neighbours on the other phases
- Watch the monitoring data for rising import on like-for-like days
- Act on smell, discolouration, buzzing or flicker at once

Reading a thermal scan (ANSI/NETA MTS criteria)

Temperature rise	Compared with	Meaning and action
1 to 3 °C	similar component, same load	Possible deficiency: investigate
4 to 15 °C	similar component, same load	Probable deficiency: repair as time permits
over 15 °C	similar component, same load	**Major discrepancy: repair immediately**
over 40 °C	ambient air	**Major discrepancy: repair immediately**

10 Conclusions

- ☑ **The joint was a fire in progress:** melted insulation and corroded copper place it at stage 3 of 5; the published cases reached fire through this same stage
- ☑ **It endangered people as well as property:** exposed live copper upstream of the RCBO, with smoke and fire to follow
- ☑ **No protective device would have stopped it:** only inspection, testing and thermal scanning find a hot joint
- ☑ **The repair removes the hazard:** provided it is confirmed by torque, insulation resistance and a thermal scan under machine load
- ☑ **The measured saving is not yet proven:** average import halved after the repair, but the 57 kW machine explains most of the day-to-day change
- ☑ **Heat is always a cost:** it is energy paid for and wasted, it grows with the square of the current, and it ends in failure if left alone
- ☑ **Every connection point must be made with care and monitored:** one conductor per terminal, correct torque, tests before handover, and a thermal scan every year under load

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11 Recommended next steps at this site

- ☑ **Inspect the RCBO terminal:** replace the RCBO if the terminal or body is discoloured, pitted or melted.
- ☑ **One conductor per terminal:** unless the maker states that the terminal accepts more.
- ☑ **Split the circuits properly:** use a busbar, a rated terminal block, or a separate breaker for each circuit.
- ☑ **Use ferrules on stranded wire and tighten to the maker's torque:** then pull-test each conductor.
- ☑ **Test the insulation:** measure the insulation resistance of every circuit before re-energising, and record the values.
- ☑ **Correct the phasing:** confirm which phase feeds each circuit and the phase sequence, then label the board.
- ☑ **Open every board by the same installer:** the second board shows the same practice
- ☑ **Scan under machine load:** record the temperature of every joint on the supply path to the 57 kW machine
- ☑ **Keep a machine log:** daily running hours, to compare energy on like-for-like days

Limits of this report

Findings on the joint are read from photographs dated 1 September 2026; the missing tests and wrong phasing are as recorded by Solar ESS on site. No photographs, thermal images or test results of the repaired joint were available, so the "after" condition is what a correct repair achieves, not a verified result. Energy figures come from six daily monitoring totals, three before and three after the repair; the repair date, system size and machine hours are inferred. Joint resistances and heat figures are worked examples from $P = I^2R$, not measurements. The demand-charge point applies only if the site's tariff includes one. Standards are cited from the secondary sources named; check the current editions before quoting them in a contract or claim.

References

1. Bangkok Metropolitan Administration fire statistics 2023–2025, reported by Matichon, 21 Dec 2025. · 2. NFPA Research, Electrical Fires Fact Sheet (2010–2014 data). · 3. London Fire Brigade findings and BS 7671:2008+A3:2015 regulation 421.1.201, reported by Professional Electrician, 11 Feb 2016. · 4. V. Babrauskas, "Fires Caused by Poor Connections", EC&M, 8 Dec 2021. · 5. Zhang et al., "Case Study on Investigation of Electrical Cabinet Fire Caused by Poor Electrical Contact", Fire 8(11):412, 2025. · 6. C. R. Miller, "General Installation Requirements, Part VI" (NEC 110.14(A)), Electrical Contractor, 15 Jul 2015. · 7. T. Korinek, "FE Investigation and Analysis of Poor Electrical Connections and Related Fire Investigation Case Studies", Journal of the National Academy of Forensic Engineers 39(1), 2022. · 8. ANSI/NETA MTS, Standard for Maintenance Testing Specifications, thermographic survey criteria (Table 100.18); see also Infraspection Institute, "Setting Temperature Limits". · 9. NFPA 70B:2023, Standard for Electrical Equipment Maintenance, section 7.4 (thermography, temperature differences documented), as summarised by FLIR. · 10. BS 7671 Table 61 / IEC 60364-6 (insulation resistance, 500 V DC, minimum 1 MΩ), as summarised by Voltimum. · 11. IEC 60479-1, Effects of current on human beings and livestock.

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